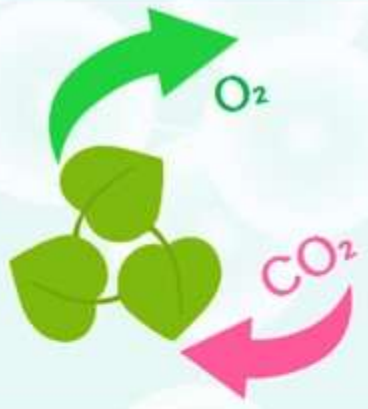


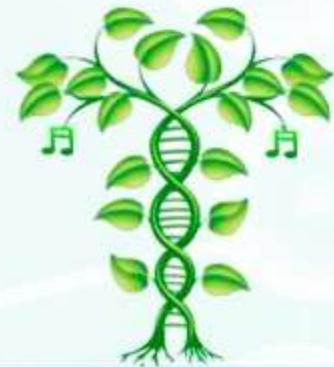


BIOLOGY



IMMUNITY

MODULE - 6



WHAT DOES HEALTH MEAN?



PHYSICALLY FIT



MENTALLY FIT

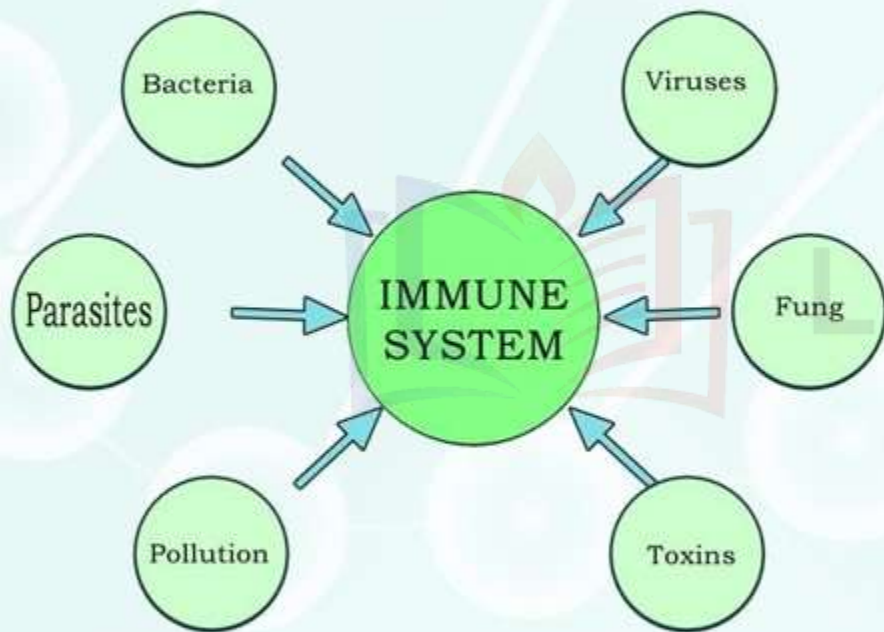


SOCIALLY FIT

Health is defined as a state of physical, mental and social well being.



HOW DOES THE BODY KEEP ITSELF SAFE AND HEALTHY?

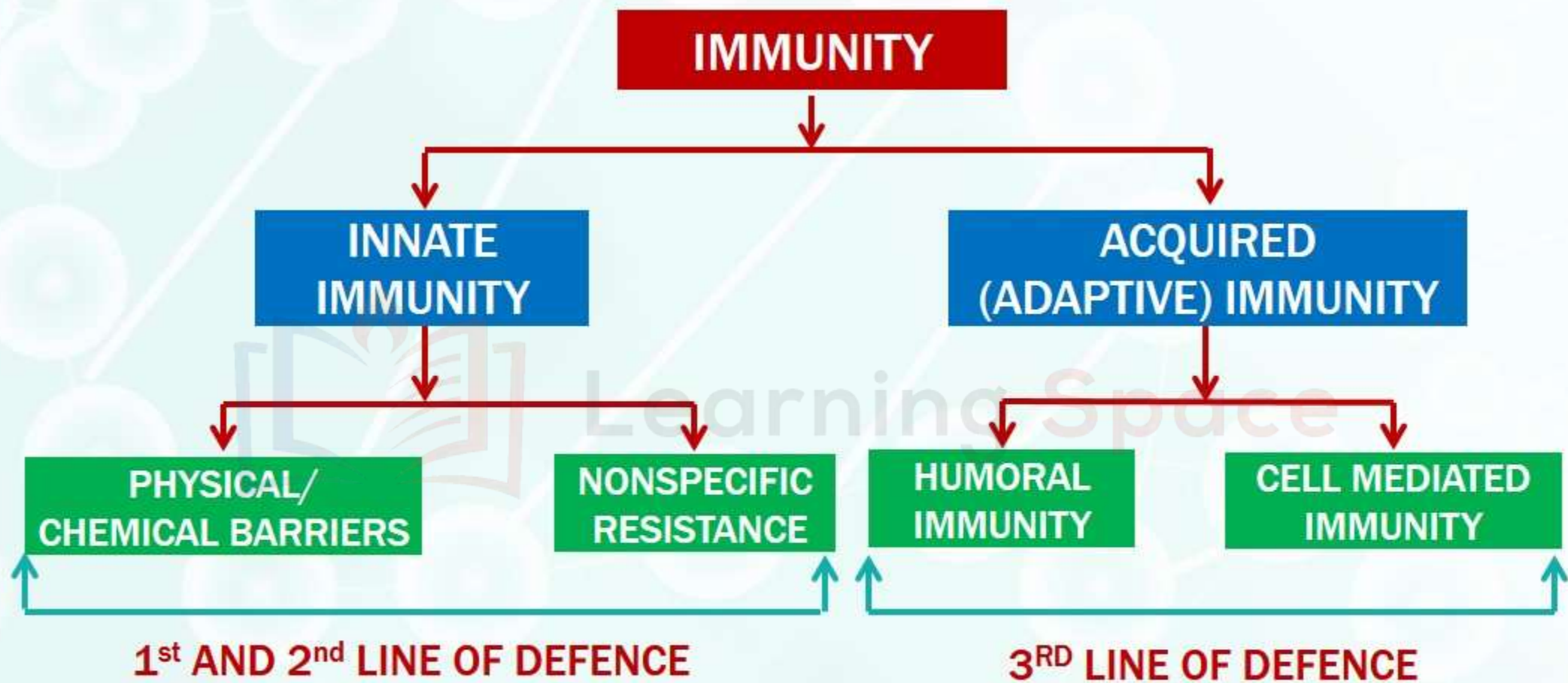


For a body to remain healthy physically it need to take care of the following:

- 1.Scavenge dead, dying body cells
- 2.Destroy abnormal (cancerous) cells
- 3.Protect from pathogens & foreign molecules: parasites, bacteria, viruses

This is where our immune system comes into action

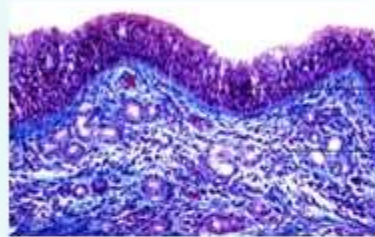




PHYSICAL BARRIERS



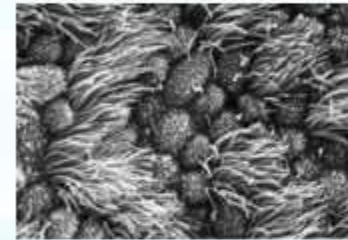
THE SKIN



MUCOUS



HAIR



CILIA



URINE



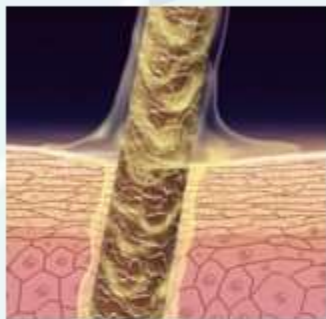
FAECES



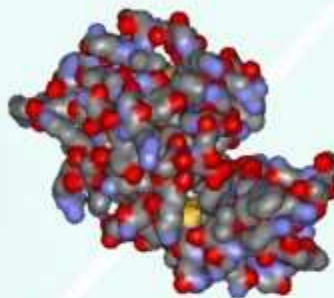
VOMITTING



CHEMICAL BARRIERS



SEBUM



LYSOZYME



SKIN ACIDITY



SALIVA



GASTRIC JUICE



HYALURONIC ACID



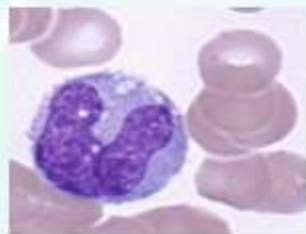
INNATE IMMUNITY

NON SPECIFIC RESISTANCE

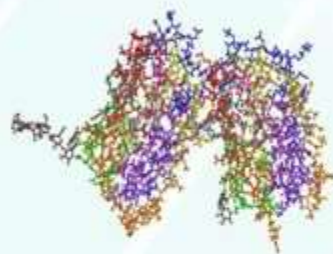
PHAGOCYTES



MACROPHAGE



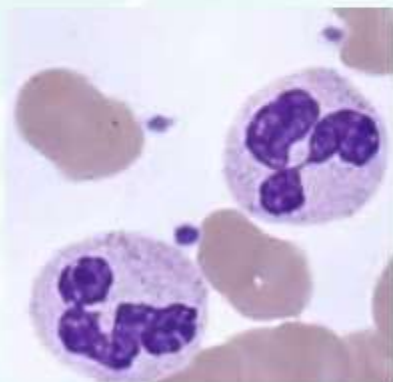
MONOCYTES



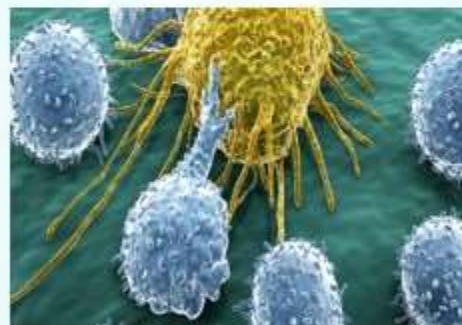
INTERFERONS



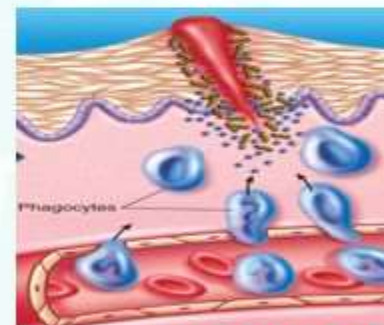
FEVER



NEUTROPHILS



**NATURAL
KILLER CELLS**



**INFLAMMATORY
RESPONSE**



CONTRAST BETWEEN THE TYPES OF IMMUNITY

IMMUNITY

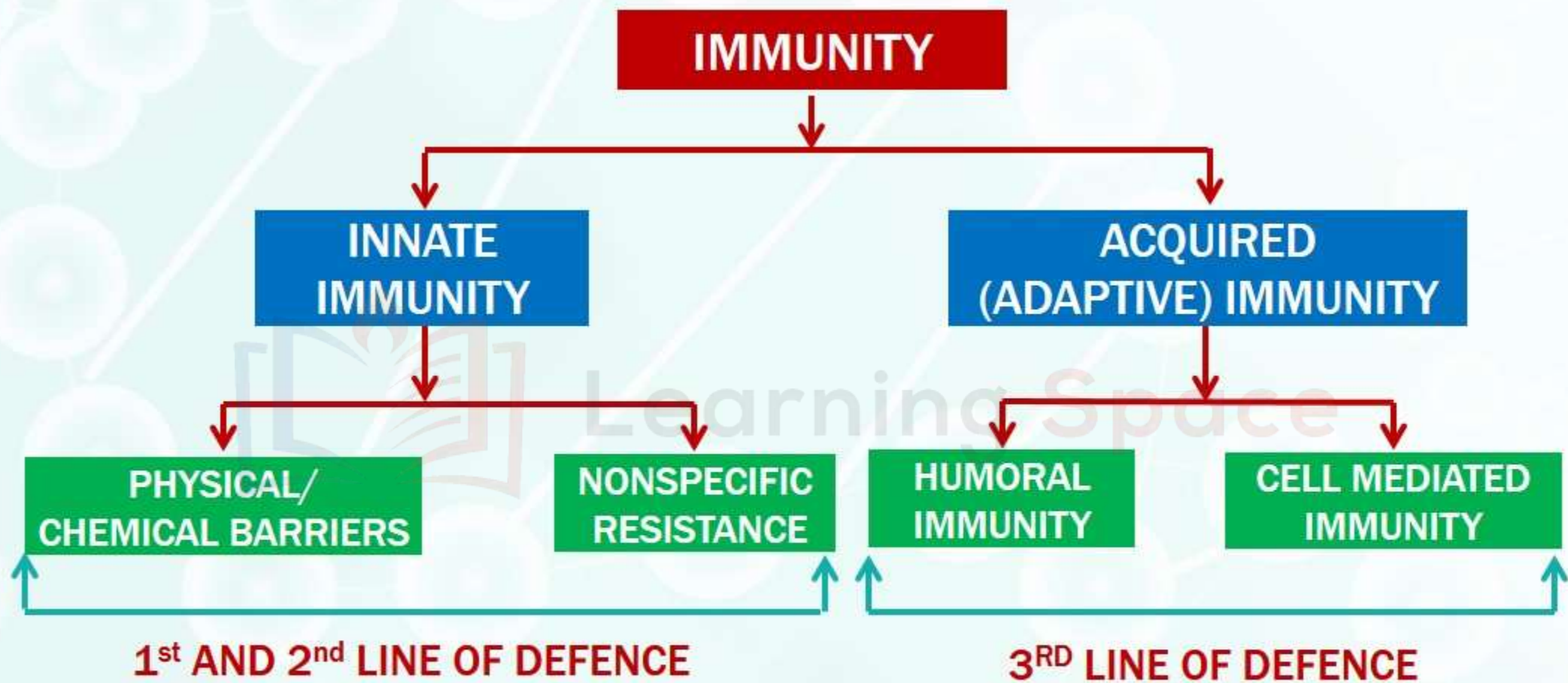
INNATE IMMUNITY

- Present in the body at all times
- Immediate response
- Specificity is limited
- Memory is absent, subsequent exposures to the same agent generate the same response

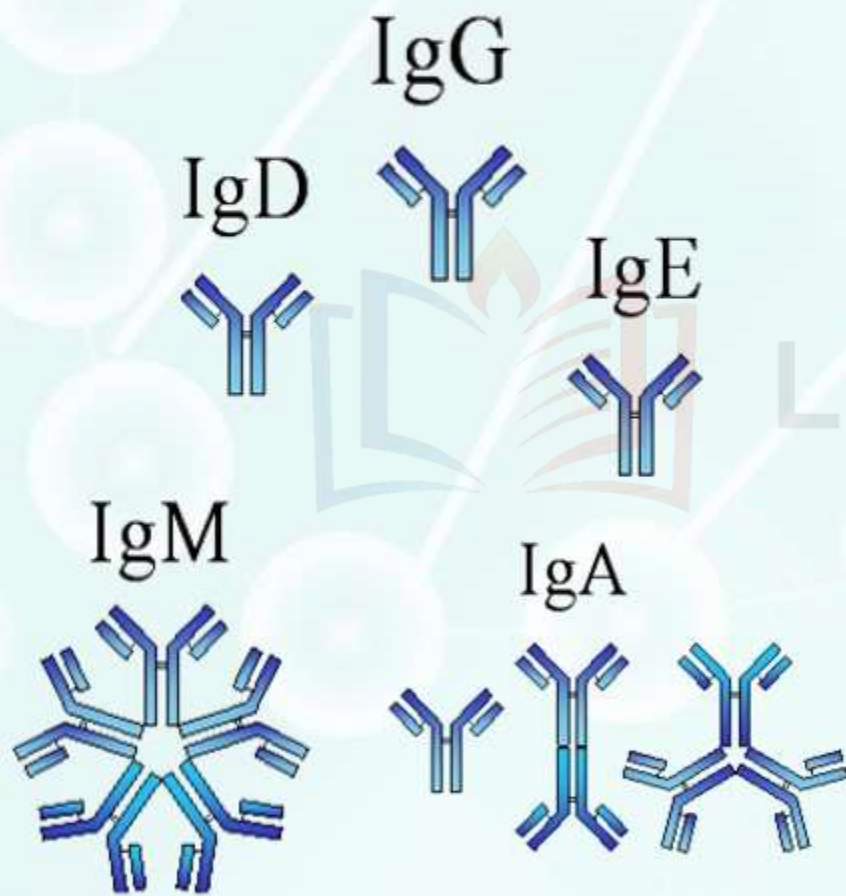
ADAPTIVE IMMUNITY

- Acquired either actively or passively
- Slow response
- High level of specificity
- Memory is present, subsequent exposures to the same agent induce an amplified response





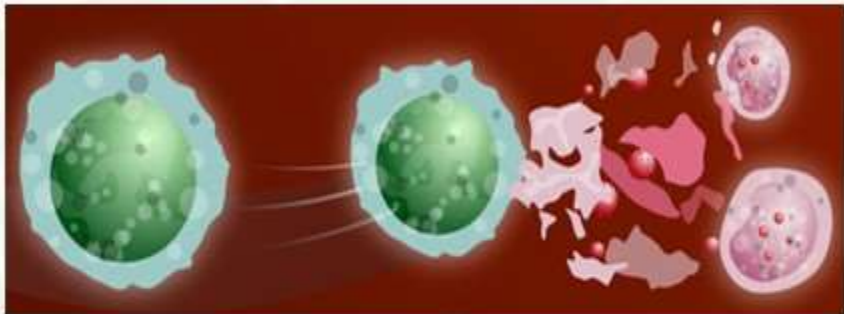
HUMORAL IMMUNITY



- **Humoral immunity is found in extracellular fluids.** Humoral immunity is so named because it involves substances found in the humors, or body fluids. It is often called antibody-mediated immunity.
- B cells, also known as B lymphocytes, are a type of white blood cell of the lymphocyte subtype. They function in the humoral immunity component of the adaptive immune system by secreting antibodies or immunoglobulins.
- B cells mature in the bone marrow and then move to the spleen and lymphoid tissues where they begin to encounter pathogens.
- **Each B cell has a unique antibody that binds with an antigen (foreign body).**
- **There are 5 isotypes or classes of immunoglobulin in mammals.**



CELL MEDIATED IMMUNITY



- Cell-mediated immunity is directed primarily at microbes that survive in phagocytes and microbes that infect non-phagocytic cells. It is most effective in removing virus-infected cells, but also participates in defending against fungi, protozoans, cancers, and intracellular bacteria.

- T-cell mediated immunity functions by activating antigen-specific cytotoxic T cells that are able to induce apoptosis in body cells, such as virus-infected cells, cells with intracellular bacteria, and cancer cells displaying tumor antigens.
- Cell mediated immunity activates macrophages and natural killer cells, enabling them to destroy pathogens.
- It stimulates cells to secrete a variety of cytokines, a type of protein cellular messenger that causes cell death.

Interferons are a type of cytokines that protects the body from viral infections



ADAPTIVE IMMUNITY...KINDS

HUMORAL IMMUNITY

- Provided by B lymphocytes
- Can recognize protein, polysaccharide, lipid and nucleic acid antigens
- Provides immunity to extracellular bacteria, viruses and toxins

CELL MEDIATED IMMUNITY

- Provided by T lymphocytes
- Can recognize only protein antigens
- Provides immunity to intracellular bacteria, viruses, fungi and protozoa.



ACQUIRED (ADAPTIVE) IMMUNITY

NATURALLY ACQUIRED

ARTIFICIALLY ACQUIRED

ACTIVE

Infection, contact with pathogen to raise natural immune cells



PASSIVE

The antibodies pass from mother to fetus via placenta or by mother's milk



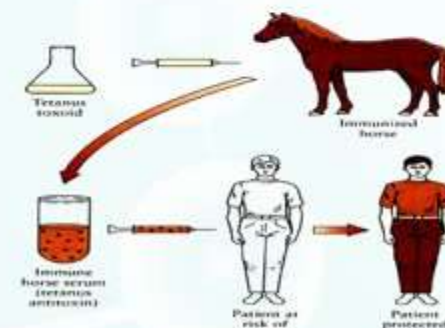
ACTIVE

Vaccines; dead or attenuated pathogens



PASSIVE

Injection of immune serum to the patient



These classification is largely for humoral immunity type



CONTRAST BETWEEN ACTIVE AND PASSIVE IMMUNITY

ACTIVE IMMUNITY

- Type of immunity produced by a reaction of a prior infection or a vaccine
- Stimulate the immune system to raise specific antibodies
- Lasts for a longer period
- Vaccination given for diseases like BCG, Polio, chicken pox, diphtheria,

PASSIVE IMMUNITY

- Antibodies produced by other animals injected into the bloodstream or through the mother
- Readymade antibodies are injected into the bloodstream
- Last for a short time
- E.g colostrum from mother's milk or prepared antibodies for tetanus, rabies, hepatitis A etc.



UNIVERSAL IMMUNIZATION PROGRAMME (UIP)



- India's Universal Immunisation Programme (U.I.P.) is one of the largest in the world in terms of quantities of vaccine used, the number of beneficiaries, the number of Immunisation session organised, the geographical spread and diversity of areas covered.
- The national policy of Immunisation of all children during the first year of life with DPT, OPV, BCG to complete the series of primary vaccination before reaching the age of 1 year was adopted in 1978. Universal Immunisation programme UIP was launched in 1985 in a phased manner.
- India was certified as polio free in March 2017.

4 new vaccines have been included in the list this year...injectable polio vaccine, vaccine against rubella, rotavirus and against adult Japanese Encephalitis.



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